A Newton-face criterion for finite-state categorical Bernoulli factories

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Abstract

Let a deterministic finite-state machine read independent samples from an unknown categorical distribution with full support and stop almost surely with a binary output. We characterize the rational target probabilities that such a machine can realize. Write the target as P/Q, where P and Q are coprime homogeneous polynomials of the same degree with rational coefficients and $0 < P < Q$ on the positive orthant. A finite-state factory exists if and only if every nonempty Newton-face polynomial of both P and $Q - P$ is strictly positive on the positive orthant. Whenever this condition holds, a fixed-block restart factory exists. The criterion is decidable by real quantifier elimination, and a successful test leads to an exact construction. The proof combines a Markov-chain tree representation of the two output probabilities, Handelman's positive-multiple theorem and the block construction of Mossel and Peres. Examples explain why the complementary numerator is essential and why powers of the coordinate sum do not exhaust the possible multipliers. No bound on decision complexity, block length or state count is asserted.

Keywords: Bernoulli factory; finite automaton; exact sampling; Newton polytope; positive polynomial multiple.

1 Introduction

A Bernoulli factory transforms samples from an unknown distribution into a bit with a prescribed probability, using the samples themselves as its source of randomness. A finite-state factory must perform this task with a fixed finite collection of memory states, although it may read arbitrarily many samples before stopping. This restriction on memory leads to a finite system of transition equations, so the output probabilities must be rational functions of the input probabilities.

For a binary source, Mossel and Peres proved that rationality is also sufficient: every rational function with rational coefficients taking values strictly between zero and one on $(0, 1)$ can be simulated by a finite-state machine [1, Theorem 2.2]. Their construction reads a block of fixed length, returns a bit for designated words, and starts again for every other word. The multivariate extension asserted in [1, Theorem 2.9] would give the same conclusion for a source with any finite number of outcomes, but Paes Leme and Schneider supplied a counterexample to that extension [2]. An additional condition on the rational target is therefore needed to characterize finite-state sampling from a categorical source.

We formulate this condition using the two output probabilities together. Suppose that $f = P/Q$ is written in reduced homogeneous form and that $0 < P < Q$ on the positive orthant. The corresponding numerators are P and $Q - P$, and each must satisfy a condition on its

Newton faces. To define these faces, take the convex hull of the polynomial's monomial exponent vectors; the terms on a nonempty face, including the whole convex hull, form its face polynomial. Theorem 2.1 states that a finite-state factory exists exactly when every such face polynomial of both numerators is strictly positive at every point with positive coordinates. This gives a finite test in terms of the target itself, and a positive answer leads to a block-restart construction.

To connect these face conditions to sampling, we use Handelman's theorem on polynomials with nonnegative-coefficient multiples [3], as presented in the account of De Angelis and Tuncel [4]. That theorem characterizes the existence of such multiples through Newton faces. We first bring a factory into this setting by obtaining nonnegative polynomial weights for both outputs from the Markov chain tree theorem [5]. Reduction of the target fraction then identifies a common multiplier of P and $Q - P$ that gives these weights. In the converse direction, the two face conditions produce separate nonnegative multipliers; their product works for both outputs, allowing the word-allocation construction of Mossel and Peres to be applied. These established results yield the complete criterion and its effective decision and construction consequences.

A related sufficient condition appears in the dice-enterprise construction of Morina, Łatuszyński, Nayar and Wendland [6, condition (5) and Theorem 3.1]. They treat rational target distributions with a finite decomposition whose factors remain strictly positive on the closed simplex, so that a closed-simplex version of Pólya's theorem applies to those factors. Their methods also encompass Bernoulli races and conditioning, which can produce targets outside that decomposition hypothesis. A comparison must therefore allow for these broader uses: a supplied pair of rational polynomial weights with nonnegative coefficients for the two outputs already yields a factory, whereas the criterion here decides whether any such representation exists and gives a terminating negative answer when none does. Section 6 gives a target outside the stated decomposition condition that nevertheless has a finite-state factory.

The necessary face conditions can also be recovered from a result in symbolic dynamics. Once a finite Markov chain is completed by resetting its outputs, the determinant and adjugate theorem of Marcus and Tuncel [7, Theorem 1] gives these conditions through its positive-polynomial matrix representation. We spell out that implication in Section 7, while using the scalar tree proof for the main argument because it keeps the connection with stopping probabilities visible and permits the finite-state and block constructions to be checked together.

The criterion concerns exact existence and decidability; it supplies no estimate for the size of the resulting machine or for the cost of the decision procedure. Throughout, termination is required only at distributions with full support.

2 The model and the criterion

Fix an alphabet $\mathcal{A} = \{1, \dots, k\}$ with $k \geq 2$, and write

$$\Delta_{k-1}^{\circ} = \left\{ p \in \mathbb{R}^k : p_i > 0 \text{ for every } i,\ \sum_{i=1}^{k} p_i = 1 \right\}.$$

The input symbols are independent with common, unknown law $p \in \Delta_{k-1}^{\circ}$. A deterministic finite-state factory consists of a finite state set, an initial state, two kinds of stopping states labelled 0 and 1, and a transition determined by the current nonstopping state and the next input symbol. No further symbol is read after a stopping state is reached. The transition rule is fixed independently of p, and there is no auxiliary random source. The machine realizes f if it stops almost surely and returns 1 with probability $f(p)$ for every $p \in \Delta_{k-1}^{\circ}$.

A **fixed-block restart factory** is a special case. Choose a length N and disjoint sets $W_0, W_1 \subseteq \mathcal{A}^N$. Read N symbols; output b when the resulting word lies in W_b, and otherwise discard the block and repeat. Its memory can store the prefix of the current block, so it uses finitely many states. Independence of successive blocks makes its eventual output probability the ratio between the one-block probability of W_1 and that of $W_0 \cup W_1$.

We state the theorem for a reduced homogeneous target

$$f = \frac{P}{Q}, \qquad P, Q \in \mathbb{Q}[x_1, \ldots, x_k], \qquad \gcd(P, Q) = 1,$$

where P and Q have the same degree and satisfy

$$0 < P(x) < Q(x) \qquad \text{for every } x \in (0, \infty)^k.$$

Put $B = Q - P$. Homogeneity makes f unchanged under positive rescaling of x, so its values on the positive orthant are determined by its restriction to the simplex.

To obtain this homogeneous form from a rational target $a(u)/b(u)$ expressed in $k-1$ independent simplex coordinates, put $s = x_1 + \cdots + x_k$ and choose d at least as large as both degrees. The polynomials $s^d a(x_1/s, \ldots, x_{k-1}/s)$ and $s^d b(x_1/s, \ldots, x_{k-1}/s)$ are homogeneous of the same degree and represent the target on the simplex. Cancelling their common homogeneous factors gives a reduced pair. The assumptions above refer to that pair; in particular, its denominator must have no zero in the domain.

For a nonzero polynomial $H = \sum_\alpha h_\alpha x^\alpha$, its support is the set of exponent vectors $\alpha \in \mathbb{Z}^k_{\geq 0}$ with $h_\alpha \neq 0$. Its **Newton polytope**, denoted $\mathcal{N}(H)$, is the convex hull of that finite set. For a nonempty face F of $\mathcal{N}(H)$, define the **face polynomial**

$$H_F(x) = \sum_{\alpha \in F} h_\alpha x^\alpha.$$

The whole polytope counts as a face, so H itself is one of these polynomials. Equivalently, a vector $w \in \mathbb{R}^k$ selects the terms whose exponents maximize $w \cdot \alpha$; every face is obtained this way, including the whole polytope by $w = 0$. Thus face selection retains the leading terms under the rescaling $x_i \mapsto t^{w_i} x_i$. It describes behaviour when the coordinates change at different rates, which ordinary positivity in the interior does not by itself control. These are faces in exponent space; they need not correspond to setting a single probability coordinate to zero.

Theorem 2.1. Under the assumptions above, the following statements are equivalent:

1. A deterministic finite-state factory realizes f on Δ°_{k-1}.
2. A deterministic fixed-block restart factory realizes f on Δ°_{k-1}.
3. Every nonempty Newton-face polynomial of P and of $B = Q - P$ is strictly positive on $(0, \infty)^k$.

For rational input coefficients, condition 3 is decidable. When it holds, an exact finite procedure constructs a fixed-block restart factory. No assertion is made about efficient complexity, bounded state count or termination at boundary points of the simplex.

The proof separates the two uses of finiteness. A finite machine first gives a ratio of two finite polynomial sums. Handelman's theorem then replaces the search over multipliers of unspecified degree by a finite list of face conditions. We prove the machine-to-polynomial step in Section 3, construct a block factory in Section 4 and make the decision procedure explicit in Section 5.

3 From finite-state sampling to polynomial weights

We write $H \geq_{\text{coeff}} 0$ when every coefficient of H is nonnegative. A nonzero polynomial with this property is strictly positive whenever all its variables are positive. Coefficientwise nonnegativity is stronger than positivity on the positive orthant, but it has a direct probabilistic interpretation: each monomial can represent the probability of a prescribed finite word.

Lemma 3.1. If a deterministic finite-state factory simulates f and $0 < f < 1$ throughout Δ_{k-1}°, there are nonzero homogeneous polynomials $A, C \in \mathbb{Z}[x_1, \ldots, x_k]$, of the same degree and with nonnegative coefficients, such that

$$f(p) = \frac{A(p)}{A(p) + C(p)} \qquad (p \in \Delta_{k-1}^{\circ}).$$

Proof. Remove unreachable states and merge all stopping states with the same output into a single state b_0 or b_1. Both are reachable, since the output probability lies strictly between zero and one. The starting state s is therefore a nonstopping state. Every retained nonstopping state has a path to one of the outputs: otherwise the factory could reach a state from which it never stops, with positive probability for each interior p.

To express the output probabilities through stationary weights, temporarily replace the stopping rule by a deterministic transition from each b_j back to s. The start reaches every retained state, and every retained state can reach an output and then the start, so the resulting finite Markov chain is irreducible. For a transition $u \to v$ from a nonstopping state, its probability is the linear polynomial

$$K_{uv}(p) = \sum_{i:\, \delta(u,i)=v} p_i,$$

where δ is the deterministic transition rule. The two reset transitions have probability one. Let π be the stationary distribution of this reset chain, for a fixed p.

To relate π to the original output probability, let $h(u)$ be the probability that the original factory eventually outputs 1 when started from u. Thus $h(b_0) = 0$, $h(b_1) = 1$ and $h(s) = f(p)$. First-step conditioning gives $Kh = h$ at every nonstopping state. At either output state, the reset gives $(Kh)(b_j) = h(s) = f(p)$. Stationarity now yields

$$0 = \sum_u \pi_u\big((Kh)(u) - h(u)\big) = \pi_{b_0} f(p) + \pi_{b_1}\big(f(p) - 1\big),$$

and hence

$$f(p) = \frac{\pi_{b_1}}{\pi_{b_0} + \pi_{b_1}}. \tag{3.1}$$

The Markov-chain tree theorem expresses these stationary weights by polynomials with nonnegative coefficients [5, p. 735]. For a state v, consider directed spanning trees oriented towards v: every state other than v has one outgoing edge, and following the edges from any state eventually reaches v. Let τ_v be the sum, over these trees, of the products of their edge weights. The theorem states that

$$\pi_v = \frac{\tau_v}{\sum_w \tau_w}.$$

For the reset chain, retain the variable x_i in each sample-reading edge weight and retain the constant one on the reset edges. Suppose there are m nonstopping states. A tree rooted at b_1 contains one outgoing edge from every nonstopping state and the reset edge from b_0; it has no outgoing edge from its root. Its weight is therefore homogeneous of degree m. The same count applies to a tree rooted at b_0. Each edge polynomial has nonnegative integer coefficients, so the two tree sums

$$A(x) = \tau_{b_1}(x), \qquad C(x) = \tau_{b_0}(x)$$

are homogeneous of degree m with nonnegative integer coefficients. Irreducibility ensures that both tree sums are nonzero. Substitution into (3.1) proves the lemma. □

The tree sums retain both output probabilities in a form whose coefficients cannot cancel. Reduction of the target fraction then identifies the polynomial factor that the two sums share.

Lemma 3.2. Under the assumptions of Theorem 2.1, a representation $P/Q = A/(A+C)$ as in Lemma 3.1 gives a nonzero polynomial $R \in \mathbb{Q}[x_1, \ldots, x_k]$ such that

$$A = PR, \qquad C = BR. \tag{3.2}$$

Proof. Equality of the fractions on the open simplex gives $AQ = P(A+C)$ there. Both sides are homogeneous of the same degree, so scaling extends the equality to the positive orthant. A polynomial that vanishes on this open set vanishes identically; thus

$$AQ = P(A+C)$$

is an identity in $\mathbb{Q}[x_1, \ldots, x_k]$. Since P and Q are relatively prime, unique factorization implies $P \mid A$. Write $A = PR$ and cancel the nonzero polynomial P to obtain $RQ = PR + C$, or $C = (Q-P)R$. □

The multiplier R in this necessity argument need not have nonnegative coefficients. Its products with P and B do have nonnegative coefficients, and this already forces positivity on every face.

3.1 Why a nonnegative multiple tests every face

Using the face interpretation from Section 2, for $w \in \mathbb{R}^k$ write $\mathrm{in}_w(H)$ for the sum of the terms $h_\alpha x^\alpha$ for which $w \cdot \alpha$ is maximal. Thus $\mathrm{in}_w(H) = H_F$ for the face F exposed by this weight, with $w = 0$ giving the whole polytope.

Lemma 3.3. Let $H \in \mathbb{R}[x_1, \ldots, x_k]$ be strictly positive on $(0,\infty)^k$. If HR is a nonzero polynomial with nonnegative coefficients for some polynomial R, then every nonempty face polynomial of H is strictly positive on $(0,\infty)^k$.

Proof. Fix a face F exposed by w, and let $M = \max\{w \cdot \alpha : h_\alpha \neq 0\}$. For each $x \in (0,\infty)^k$, rescaling the variables gives

$$H_F(x) = \lim_{t\to\infty} e^{-Mt} H(x_1 e^{w_1 t}, \ldots, x_k e^{w_k t}) \geq 0. \tag{3.3}$$

Terms outside F disappear in this limit because their w-weights are smaller than M. Positivity of H therefore gives nonnegativity of H_F, but it does not by itself exclude a zero.

The product HR excludes that remaining possibility. Highest-weight parts multiply according to

$$\mathrm{in}_w(HR) = \mathrm{in}_w(H)\,\mathrm{in}_w(R) = H_F\,\mathrm{in}_w(R). \tag{3.4}$$

Indeed, terms below the highest weight cannot contribute to the highest-weight part of the product, and the product of the two highest-weight parts is nonzero because the polynomial ring has no zero divisors. The left-hand side of (3.4) is a nonzero polynomial with nonnegative coefficients. It is consequently strictly positive at every $x \in (0,\infty)^k$, which forces $H_F(x) \neq 0$. Combined with (3.3), this gives $H_F(x) > 0$. □

Apply Lemma 3.3 first to $H = P$ and then to $H = B$, using the products in (3.2). This proves the implication from condition 1 to condition 3 of Theorem 2.1. The argument is the elementary necessity direction of the positive-multiple theorem; its role here is to identify the two polynomials to which that established condition must be applied.

4 From face positivity to a block factory

For the converse, we use Handelman's positive-multiple theorem as presented by De Angelis and Tuncel [4, Handelman's Theorem; author version, pp. 1–4]. If every nonempty face polynomial of a polynomial H is strictly positive on the positive orthant, and G has strictly positive coefficients on exactly the support of H, then

$$G^n H \geq_{\mathrm{coeff}} 0$$

for all sufficiently large integers n. Applying this form in the rational polynomial ring, choose

$$G_H = \sum_{\alpha \in \mathrm{supp}(H)} x^\alpha.$$

Its powers have integer coefficients, and when H is homogeneous, G_H is homogeneous of the same degree. The resulting multiples therefore retain the rational coefficients and homogeneity needed to construct the word probabilities.

Lemma 4.1. If condition 3 of Theorem 2.1 holds, there are nonzero homogeneous polynomials $A, C \in \mathbb{Z}[x_1, \dots, x_k]$, of the same degree and with nonnegative coefficients, such that

$$\frac{P}{Q} = \frac{A}{A + C}.$$

Proof. Apply the preceding theorem separately to P and B. There are nonnegative integers n_P, n_B for which

$$PG_P^{n_P} \geq_{\mathrm{coeff}} 0, \qquad BG_B^{n_B} \geq_{\mathrm{coeff}} 0.$$

Write $U = G_P^{n_P}$ and $V = G_B^{n_B}$. Because these multipliers themselves have nonnegative integer coefficients, their product works for both outputs:

$$P(UV) = (PU)V \geq_{\mathrm{coeff}} 0, \qquad B(UV) = (BV)U \geq_{\mathrm{coeff}} 0.$$

Choose a positive integer D that clears the coefficients' denominators and set

$$A = DPUV, \qquad C = DBUV.$$

They are nonzero homogeneous integer polynomials of the same degree. Finally, $P + B = Q$ gives $A/(A + C) = P/Q$. $\square$

These polynomial weights determine the relative probabilities of the two outputs. To turn them into a deterministic factory, we must also realize their integer coefficients as numbers of distinct words. A monomial x^α of degree L is the probability of any particular length-L word with symbol-count vector α; its coefficient may nevertheless exceed the number of such words. The following padding construction supplies enough words without changing the output ratio. It is the categorical version of the balanced-word construction in Mossel–Peres [1, Proposition 2.5].

Lemma 4.2. Let A, C be nonzero homogeneous polynomials of the same degree with nonnegative rational coefficients. Then a deterministic fixed-block restart factory simulates $A/(A + C)$ on Δ_{k-1}°.

Proof. Multiply A and C by a common positive integer so that all coefficients are integers, and write

$$A = \sum_{|\alpha| = L} a_\alpha x^\alpha, \qquad C = \sum_{|\alpha| = L} c_\alpha x^\alpha.$$

Choose $r \geq 1$ such that

$$\binom{2r}{r} \geq \max_\alpha (a_\alpha + c_\alpha),$$

and set $N = L + 2r$. For each α with $a_\alpha + c_\alpha > 0$, fix a word u_α of length L containing exactly α_i copies of symbol i. Append to u_α each binary word of length $2r$ containing r copies of symbol 1 and r copies of symbol 2. This gives $\binom{2r}{r}$ distinct length-N words, each with probability

$$p^\alpha p_1^r p_2^r.$$

For this α, assign a_α of the resulting words to $\mathcal{W}_1$ and c_α different words to $\mathcal{W}_0$. Word sets chosen for different α cannot intersect, because their full symbol-count vectors differ. Thus $\mathcal{W}_0$ and $\mathcal{W}_1$ are disjoint.

Read length-N blocks, stop on a word in one of the assigned sets, and restart otherwise. The probabilities of output 1 and output 0 in a single block are, respectively,

$$q_1(p) = p_1^r p_2^r A(p), \qquad q_0(p) = p_1^r p_2^r C(p).$$

Their sum $q(p)$ lies in $(0, 1]$: it is positive because p has positive coordinates and A, C are nonzero, and it is at most one because the word sets are disjoint. Independence of successive input blocks gives

$$\mathbb{P}_p(\text{eventual output } 1) = \sum_{j=0}^{\infty} (1 - q(p))^j q_1(p) = \frac{q_1(p)}{q(p)} = \frac{A(p)}{A(p) + C(p)}.$$

The probability of surviving j complete blocks is $(1 - q(p))^j$, which tends to zero. The construction therefore stops almost surely for every interior p. It is a finite-state machine: store the current prefix up to length $N - 1$, and on the last symbol either enter the designated output state or return to the starting state. Every transition uses exactly the next source symbol. □

Lemmas 4.1 and 4.2 prove that condition 3 implies condition 2. Every fixed-block restart factory is a finite-state factory, so condition 2 implies condition 1. This completes the equivalence in Theorem 2.1.

5 Decision and construction

The criterion replaces an existence question over machines of unrestricted finite size by finitely many polynomial inequalities. Given the rational coefficients of P, Q, compute $B = Q - P$ and enumerate the faces of the two Newton polytopes. This is an effective finite operation on their integer exponent vectors. For each face polynomial H_F, test the first-order sentence

$$\forall x_1 \cdots \forall x_k \left[\left(\bigwedge_{i=1}^{k} x_i > 0 \right) \Longrightarrow H_F(x_1, \ldots, x_k) > 0 \right]. \tag{5.1}$$

Effective quantifier elimination over the real numbers decides each sentence [8, Chapter 14]. If any one is false, the necessity argument rules out a finite-state factory. If every sentence is true, the converse proves existence and also supplies a terminating construction search.

For that search, form G_P and G_B as the sums of their support monomials. Test the coefficients of PG_P^n for $n = 0, 1, 2, \ldots$ until all are nonnegative, and do the same for BG_B^n. The positive-multiple theorem guarantees that both searches terminate after the face tests have succeeded. All coefficient calculations use exact rational arithmetic. The common multiplier, denominator clearing and finite word allocation of Section 4 then give the required machine.

The negative decision comes from (5.1), not from waiting for an unsuccessful multiplier search. Neither the degree at which the search succeeds nor the number of states produced here is controlled by a useful complexity bound. For every fixed interior parameter, however, the constructed factory has a geometric tail in its number of blocks, as the calculation in Lemma 4.2 shows. No uniform estimate as p approaches the boundary is required by the theorem.

6 Examples and the boundary of the sufficient constructions

6.1 A negative coefficient that a multiplier removes

Consider a source with three letters X, Y, Z and probabilities x, y, z, respectively. Set

$$P = x^2 - xy + y^2, \qquad Q = P + z^2.$$

These polynomials are coprime, since a common factor would divide both P and z^2. They satisfy $0 < P < Q$ on the positive orthant. The Newton polytope of P is the segment joining $(2,0,0)$ to $(0,2,0)$. Its endpoint face polynomials are x^2 and y^2, and its whole-face polynomial is

$$P = (x-y)^2 + xy > 0.$$

The complementary numerator is $B = z^2$, so its only face polynomial is also strictly positive. Theorem 2.1 therefore gives a finite-state factory despite the negative coefficient of xy in P. Here the multiplier and the factory are small enough to write directly:

$$A = (x+y)P = x^3 + y^3, \qquad C = (x+y)B = xz^2 + yz^2.$$

Read blocks of three symbols, output 1 on XXX or YYY, output 0 on XZZ or YZZ, and restart otherwise. The one-block output probabilities are A and C. Their sum is positive in the open simplex, so repeated blocks stop almost surely and yield

$$\frac{A}{A+C} = \frac{P}{P+z^2}.$$

Figure 1 displays the entire procedure. The 23 remaining words are restart words; their number does not imply that they are equiprobable. The example shows why coefficient signs in a single chosen numerator cannot decide the existence question.

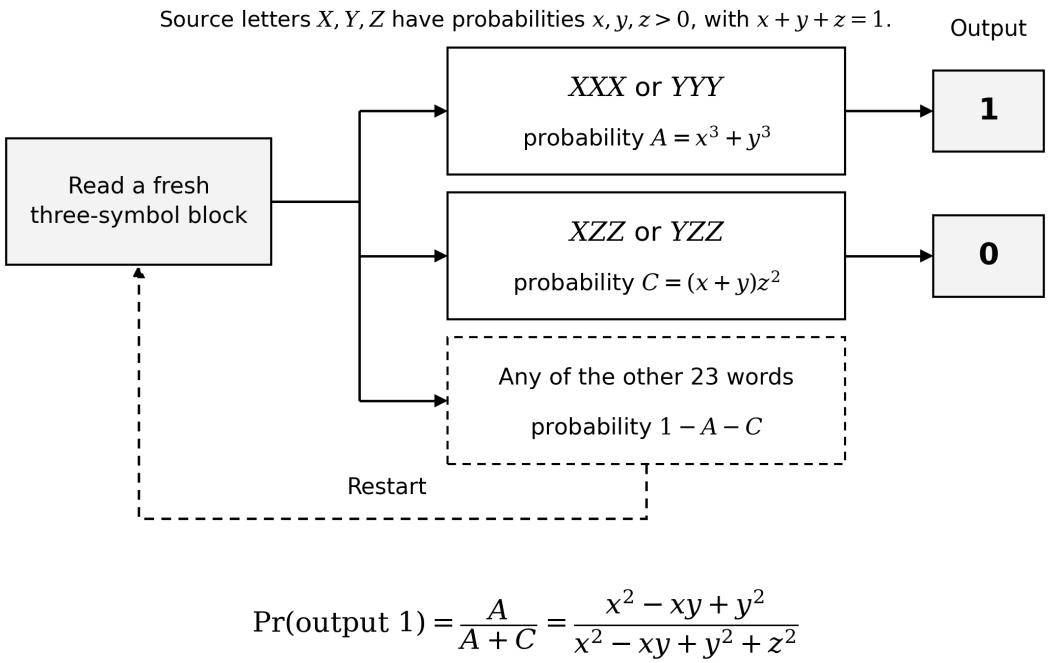


Figure 1: A block factory for $(x^2 - xy + y^2)/(x^2 - xy + y^2 + z^2)$. The letters X, Y, Z have probabilities x, y, z. The four stopping words have total probability $x^3 + y^3 + (x+y)z^2$; every other three-letter word restarts the procedure. The common factor $x + y$ cancels in the eventual output ratio.

6.2 A complementary face that rules out finite-state sampling

Now consider

$$f = \frac{4xy}{(x+y+z)^2}.$$

The numerator and denominator have nonnegative coefficients, and $0 < f < 1$ on the positive orthant. Nevertheless, the complementary numerator is

$$B = (x + y + z)^2 - 4xy = (x - y)^2 + 2xz + 2yz + z^2.$$

The face selected by the largest value of $\alpha_x + \alpha_y$, equivalently by $\alpha_z = 0$, has polynomial

$$B_F = x^2 - 2xy + y^2 = (x - y)^2.$$

It vanishes when $x = y > 0$, even with $z > 0$. This face therefore fails the strict positivity required by Theorem 2.1, and no finite-state factory realizes f on the whole open simplex. The zero belongs to the face polynomial; the full complementary numerator remains positive in that domain. Figure 2 compares this face with the positive segment polynomial in the preceding example.

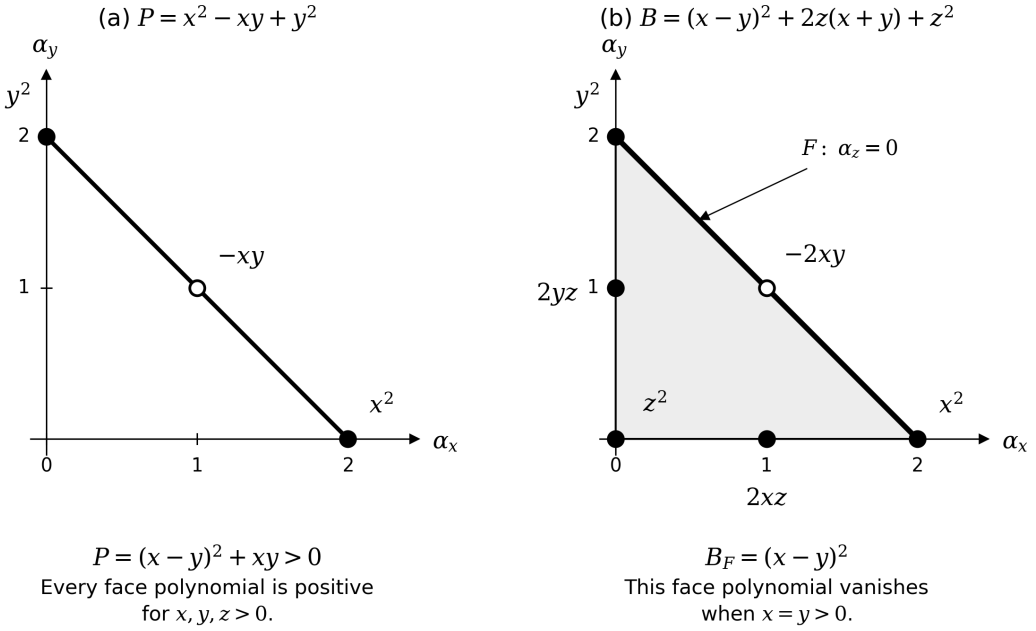


Figure 2: Newton faces for two different target functions, in exponent coordinates. Since every monomial has degree two, the plotted coordinates (α_x, α_y) determine $\alpha_z = 2 - \alpha_x - \alpha_y$. In the left panel, the complete segment polynomial $x^2 - xy + y^2$ is positive for $x, y > 0$, although its middle coefficient is negative. For the target $4xy/(x + y + z)^2$ in the right panel, the edge $\alpha_z = 0$ of the complement's Newton triangle carries $(x - y)^2$, which vanishes at $x = y > 0$. Filled markers indicate positive coefficients and hollow markers negative coefficients, also recorded by the signed labels. Points represent exponents, not input probabilities.

The complement is necessary because a stopping machine has to allocate nonnegative probabilities to both outputs. Positivity of the numerator and denominator coefficients alone does not ensure that their difference has a nonnegative multiple. This requirement is already present in the coefficient inequality of Mossel–Peres [1, Proposition 2.5]; the face criterion states it directly for the reduced rational target.

6.3 Why coordinate-sum powers are insufficient

For the first example, multiplication by $x + y$ removes the negative coefficient. Multiplication by a power of $s = x + y + z$ never does: for every $n \geq 0$, the coefficient of xyz^n in $s^n(x^2 - xy + y^2)$

is -1. The positive terms x^2 and y^2 cannot contribute to that monomial. A failed search using only powers of s would therefore continue indefinitely even though a three-symbol block factory is available.

This distinction between coordinate-sum powers and other multipliers is already visible in Paes Leme's tutorial [9, Example 2.3], which credits a suggestion by Schneider. Its example is

$$g = 3xy^2 + y^3 - 3xyz + 3xz^2 + z^3.$$

The polynomial is positive on the positive orthant, while the coefficient of $x^{n+1}yz$ in $s^n g$ is always -3. It nevertheless has the nonnegative-coefficient multiple

$$(y + z)g = (y^3 + z^3)(3x + y + z).$$

The failure of one prescribed multiplier family thus does not decide finite-state realizability. The Newton-face criterion tests existence of the required multiples without choosing that restricted family in advance.

6.4 Separation from the final dice-enterprise hypothesis

The target $f = P/(P + z^2)$ of Section 6.1 has an explicit block factory, but it does not satisfy condition (5) in [6, p. 333]. That condition represents each target coordinate as a finite sum of monomials times factors that remain strictly between zero and one on the closed simplex. The proof treats these factors as rational functions regular there, since it applies its Lemma B.2 to them [6, pp. 349–351].

To check this independently of any chosen numerator and denominator, approach the vertex $(0, 0, 1)$ along

$$x = ta, \qquad y = tb, \qquad z = 1 - t(a + b), \qquad a, b > 0.$$

As $t \downarrow 0$, the target satisfies

$$\frac{f(ta, tb, 1 - t(a + b))}{t^2} \longrightarrow a^2 - ab + b^2.$$

Suppose it had a finite decomposition into terms $x^\alpha y^\beta z^\gamma h(x, y, z)$, with nonnegative integer exponents and each h continuous and strictly positive at the vertex. The leading term of such a summand is

$$t^{\alpha+\beta} a^\alpha b^\beta h(0, 0, 1).$$

All leading coefficients are positive. A term with $\alpha + \beta < 2$ would contradict the order-two vanishing of f, because no other summand could cancel it. Terms with $\alpha + \beta > 2$ disappear after division by t^2. Consequently a quadratic polynomial with nonnegative coefficients would equal $a^2 - ab + b^2$ for every $a, b > 0$, contradicting the coefficient -1 of ab.

This argument separates the target from the stated sufficient hypothesis of [6, Theorem 3.1]. It does not exclude using the paper's Bernoulli races, conditioning or other preprocessing to construct a sampler. Those operations, as well as a supplied nonnegative polynomial representation, remain available. The additional content of Theorem 2.1 is the complete test for such a representation and for finite-state realizability.

6.5 Recovery of the binary-source classification

When $k = 2$, the Newton polytope of a homogeneous polynomial is a point or a segment. A polynomial positive on the open quadrant has positive endpoint coefficients: with one coordinate fixed at one, the lowest- and highest-degree terms in the other coordinate determine its sign near zero and infinity, respectively. Their coefficients must therefore be nonnegative, and they are nonzero by the definition of support and hence positive. Its only other nonempty face is the entire segment, whose polynomial is positive by assumption.

Apply this observation to P and B. Every reduced rational probability in the binary-source model satisfies the face condition, recovering the rational-function classification of Mossel and Peres [1, Theorem 2.2]. For three or more source outcomes, a proper face can contain several monomials; Section 6.2 shows how cancellation within such a face changes the answer.

7 Relation to positive-polynomial matrices and scope

The necessity direction also follows from an established theorem in symbolic dynamics. A polynomial whose every nonempty face polynomial is strictly positive on the positive orthant is called a Handelman polynomial in [7, Section 2]. Marcus–Tuncel [7, Theorem 1] treats irreducible matrices whose nonzero entries are such polynomials and whose Perron eigenvalue is itself a polynomial. For a proper principal submatrix, it shows that the characteristic determinant evaluated at that eigenvalue, and every nonzero entry of the corresponding adjugate, are Handelman polynomials.

For the factory, put $s = x_1 + \cdots + x_k$, let T collect the sample-reading transition weights between its retained nonstopping states, and now let b_0, b_1 denote the two exit columns. As in Lemma 3.1, add the outputs back with transitions to the starting state. Assigning each reset edge weight s gives every row the same sum s; the resulting matrix M is irreducible and all its entries have nonnegative coefficients. Consequently $M(x)\mathbf{1} = s(x)\mathbf{1}$ for positive x, so its Perron eigenvalue is the polynomial s. Its nonzero entries satisfy the face condition automatically, and T is a proper principal submatrix. To recover the output probabilities from this matrix, define

$$D = \det(sI - T), \qquad N_j = e_{\text{start}}^T \operatorname{adj}(sI - T) b_j.$$

The cited theorem makes D and the nonzero adjugate entries Handelman polynomials. Sums and products of such polynomials have the same property: Handelman's theorem supplies nonnegative multipliers, whose products give a common multiplier as in Lemma 4.1. The nonzero entries of b_j also have this property, so each nonzero N_j does. Both N_j are nonzero because both outputs are reachable. On the simplex, $s = 1$, so the absorption equations give output-1 probability N_1/D. Moreover,

$$b_0 + b_1 = (sI - T)\mathbf{1}$$

gives $N_0 + N_1 = D$.

The same reduction argument as in Lemma 3.2 gives $N_1 = PR$ and $N_0 = BR$ for a common polynomial R. A nonnegative-coefficient multiple of N_1 is then a nonnegative-coefficient multiple of P, and likewise for N_0 and B. Lemma 3.3 yields the two required face conditions. Thus the symbolic-dynamics theorem already supplies the relevant algebraic necessity after this conversion, while the tree argument in Section 3 gives a direct probabilistic representation with nonnegative coefficients. The main equivalence can therefore be proved using the stated tree and positive-multiple theorems, without the additional matrix machinery.

The resulting classification concerns a fixed finite memory space, rational target coefficients and pointwise almost-sure stopping throughout the open simplex. Rational coefficients permit exact polynomial operations and effective real quantifier elimination; full support makes every selected stopping word have positive probability. The construction can require large multipliers and long blocks, and its stopping probability may approach zero near the boundary. Finding useful complexity bounds is a separate quantitative problem.

Declarations

Data and code. The results are theoretical. No experimental dataset or numerical computation is used in the proofs. The figures depict the explicitly stated constructions and exponent sets; their generation code accompanies the manuscript source.

AI assistance. AI assistance was used during proof development, source comparison and manuscript preparation. The author is responsible for the mathematical content, references and final text.

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